

Aleksei Liuliakov

PhD in Machine Learning, dissertation submitted September 2026, defence expected December 2026

- **DGPO** (IJCNN 2026): RL-steered discrete **graph diffusion** for **neural architecture generation**; matches the NAS-Bench-201 optima on all three tasks (CIFAR-10 91.61%, CIFAR-100 73.49%, ImageNet-16-120 46.77%; by benchmark lookup); pretrained on 7% of NAS-Bench-101, the best generated architecture lands within 0.32 pp of the full-data model. [\[arXiv\]](#) [\[code\]](#)
- **Anomaly detection on dynamic graphs**: TGN-SVDD reaches ROC-AUC 0.994–0.999 vs 0.268–0.690 for vanilla TGN on CIC-IDS2017 (ICANN 2023); probabilistic RTGN-SVDD raises ROC-AUC 0.574 → 0.844 and 0.561 → 0.880 under 50% label contamination vs the deterministic model (ESANN 2024).
- **Multi-objective AutoML** (TPOT, Optuna, NSGA-II): Pareto fronts between accuracy and sparsity show that different architectures are optimal at different sparsity levels for intrusion detection (UNSW-NB15, NSL-KDD) and outlier detection (Applied Soft Computing, 2023).

Experience

- Nov 2020 – Present **Doctoral Researcher – Machine Learning**, *Bielefeld University*, Bielefeld, Germany
- DGPO: extended RL-steered discrete graph diffusion (DiGress + REINFORCE-style policy optimization) from undirected molecular graphs to DAGs via topological node ordering and positional encoding; 94.50% on NAS-Bench-101 (benchmark lookup); multi-objective MO-DGPO reaches Pareto hypervolume 0.9901 on NAS-Bench-201, matching the union of three single-task runs.
 - Failure-mode check: bidirectional-control ablation, where inverse reward drives accuracy to 9.5% (chance), validating directional control of the generator.
 - Ran multi-seed training and evaluation campaigns on SLURM GPU clusters (3 seeds; single-GPU, ~115 GPU-hours for DGPO on one A40).
 - Engineering: public research repos **DGPO** (MIT, Makefile one-command reproduction, pinned requirements) and **rtgn_svdd** (pyproject, pytest tests, docs).
 - Published 6 papers (5 as first author); built AI-augmented research workflows (Claude Code, Codex) to automate experimental pipelines, from implementation through systematic evaluation.
- Oct 2023 – Sep 2025 **Tutor & Supervisor**, *Bielefeld University*
- Applied Optimization tutorials (coordinated fellow tutors); sole seminar instructor, 'Implications of the AI Act in Machine Learning'.
- Sep 2011 – Jul 2018 **Founder & CEO – Baltik Bread**, *Bakery Manufacturing & Mobile Retail*, Novosibirsk, Russia
- Co-founded and scaled a dual manufacturing + retail operation from zero; 500k-600k EUR investment; full ROI within 4 years.
 - Grew to ~1M EUR annual revenue, ~2,500 units/day across ~40 SKUs with 90%+ sell-through; built and managed a 25-person team.
 - Data-driven location optimization using empirical sales analysis; built competitive advantage through systematic testing.

Education

- Nov 2020 – Sep 2026 **PhD (Dr. rer. nat.) in Machine Learning**, *Bielefeld University*, Bielefeld, Germany
- Dissertation submitted Sep 2026, defence expected Dec 2026. Supervisor: Prof. Barbara Hammer.
- Oct 2018 – Mar 2021 **M.Sc. Data Science**, *Bielefeld University*, Bielefeld, Germany
- Sep 2007 – Jun 2010 **M.Sc. Econometrics & Quantitative Economics**, *Novosibirsk State University*, Novosibirsk, Russia
- Sep 2003 – Jun 2007 **B.Sc. Physics**, *Novosibirsk State University*, Novosibirsk, Russia

Publications

- **A. Liuliakov**, L. Hermes, and B. Hammer. “DGPO: RL-Steered Graph Diffusion for Neural Architecture Generation.” In: *IJCNN 2026 (WCCI)*. [arXiv] [code]
- **A. Liuliakov**, A. Schulz, L. Hermes, and B. Hammer. “Noise Robust One-Class Intrusion Detection on Dynamic Graphs.” In: *ESANN 2024*. [arXiv] [code]
- **A. Liuliakov**, A. Schulz, L. Hermes, and B. Hammer. “One-Class Intrusion Detection with Dynamic Graphs.” In: *ICANN 2023*. [arXiv] [code]
- L. Hermes, **A. Liuliakov**, and M. Schilling. “Graph Learning by Dynamic Sampling.” In: *IJCNN 2023*.
- **A. Liuliakov**, L. Hermes, and B. Hammer. “AutoML Technologies for the Identification of Sparse Classification and Outlier Detection Models.” In: *Applied Soft Computing*, 2023.
- **A. Liuliakov** and B. Hammer. “AutoML Technologies for the Identification of Sparse Models.” In: *IDEAL 2021*.

Service

Session chair	‘AI for Finance and Economics’, IJCNN 2026 (IEEE WCCI, Maastricht)
Peer reviewing	ESANN 2024 (1), IJCNN 2024 (1), ESANN 2026 (2), IJCNN 2026 (2) papers
Teaching	Sole seminar instructor, ‘Implications of the AI Act in Machine Learning’, Bielefeld University, SoSe 2024 and SoSe 2025

Skills

ML & AI	Generative AI, Diffusion Models, Reinforcement Learning, Graph Neural Networks, Deep Learning, AutoML, Neural Architecture Search, Anomaly Detection, OOD Robustness, Probabilistic Modeling
Programming	Python, PyTorch, PyTorch Geometric, PyTorch Lightning, Optuna, scikit-learn, NumPy, Pandas, R, JavaScript, HTML, CSS
Platforms	Linux clusters, Slurm, Docker, W&B, Git, GitHub, VS Code
Engineering	Public research repos: DGPO (MIT, Makefile one-command reproduction, pinned requirements); rtgn_svdd (pyproject, pytest tests, docs)
AI Tools	Claude Code, OpenAI Codex, GitHub Copilot; agentic workflows for experiment automation (DiGress/DGPO training, RL fine-tuning, NAS evaluation) and GPU cluster deployment via SLURM
Research Methods	Mathematical Modeling, Statistics, Applied Optimization, Experimental Design, Scientific Writing

Languages

English	Full professional proficiency (C1)
German	Limited working proficiency (B1)
Russian	Native
French	Elementary (A1)